Synthetic Biological Membrane

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For long duration human space flight missions, reliability is a critical factor. It impacts both safety and cost. This is particularly true for spacecraft life support systems. This paper describes the development of a biomimetic technology called the Synthetic Biological Membrane (SBM). The SBM improves the reliability of separation membranes by mimicking the regenerative function of biological membranes. It uses genetically engineered organisms to regenerate a protective fatty acid layer on the surface of a lipid membrane, engineered to provide a sacrificial protective surface on the feed side of the membrane. This surface protects the membrane from mineral scale, organic fouling, and oxidative damage. The SBM is configured for use in a forward osmosis (FO) water purification system. In this configuration the engineered organisms are grown in the recirculating osmotic agent and the fatty acids they produce diffuse across the lipid forward osmosis membrane into the feed.

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Nomenclature

SBM	=	synthetic biological membrane
ISS	=	international space station
FO	=	forward osmosis
OA	=	osmotic agent
FA	=	fatty acid
FALCs	=	fatty alcohols
SOA	=	state-of-the-art
KPP	=	key performance parameters
Fy	=	*Y* component of the resultant pressure force acting on the vehicle
f, g	=	generic functions

I. Introduction

The project described herein addresses the development of a new type of biomaterial that will improve the reliability of NASA's next generation life support systems. Its justification emanates from the lessons learned during operation of the International Space Station (ISS). Those lessons demonstrated that reliability is a key limitation to current ISS technologies. ISS is a complicated machine and like all machines it breaks down. For example, the ISS life support system, a mission critical system, requires maintenance or repair every 90 days. For a typical Mars mission, that could last three years or longer, this limitation would affect both mission safety and cost. NASA's current approach to address reliability is to require triple redundancy for all mission critical subsystems. This requirement increases up-mass and has a significant impact on launch cost. The objective of the project described here is to improve subsystem reliability by integrating biological functions with mechanical systems to create a new type of biomechanical subsystem.

The human heart, lungs, and digestive system have an expected life greater than 50 years with minimal to no maintenance. This is because biological systems have undergone billions of years of evolutionary improvements. Mechanical spacecraft pumps, air handling systems, and water treatment systems have expected lives of moths to years and require extensive maintenance to operate. This is because they have undergone less than 100 years of "evolutionary improvements. This project will transfer lessons learned from the evolution of the human small intestine to develop a regenerative membrane. This biomimetic membrane will be used in manned spacecraft as part of the water recycling system previously developed by NASA.

Membranes are a vital component of many biological processes, and are used extensively in modern spacecraft because of their low mass, power, and volume requirements. They are traditionally used for such diverse applications as space suits, chemical separations, sensors and structural components such as vapor barriers. However, the use of membranes in space applications is limited by their relative fragility and short operational life. Conventional, man-made, polymer membranes are susceptible to chemical, physical, biological and radiation-induced damage. For long duration space flight missions this results in an increased resupply mass and an increase in crew time required for maintenance, repair and replacement of these membranes.

Accumulation of unwanted materials can cause membrane fouling, that leads to a decrease in membrane permeability. The fouling could be reversible if the layer of foulants is formed on top of the membrane, or irreversible, if the pores of the membrane get blocked by the foulants [1.] Fouling is typically caused by mineral scale deposits or organic foulants.

The organic fouling of a membrane is caused by a deposition of biopolymers. Previous studies have demonstrated that the major percentage of those polymers are proteins and polysaccharides [2]. more specifically polysaccharides and other non-setteable organic matter with a molecular weight larger than 120 000 Da [3].Biological precipitation can be another contribution to fouling. The biopolymers contain ionizable groups (COO^-, CO_3^{2-}, SO_4^{2-}, PO_4^{3-} and OH^-) that are easily captured by metal ions. Metal ions play a significant role in the formation of fouling layers, which can bridge the deposited cells and biopolymers and then form a dense caked layer. Additionally, a synergistic interaction among biofouling, organic fouling and inorganic fouling[4] exists creating a more significant problem.

Mineral scale composed mainly of calcium carbonate ($CaCO_3$), magnesium hydroxide ($Mg(OH)_2$), and calcium sulfate ($CaSO_4$) can cause the membrane function to decrease. [5] Calcium and magnesium carbonates tend to be deposited as off-white solids on the surfaces of membranes. This precipitation is principally caused by thermal decomposition of bicarbonate ions but also happens in cases where the carbonate ion is concentrated above its saturation limit [6]. The resulting build-up of scale restricts the flux of water across membranes. The damage caused by calcium carbonate deposits varies depending on the crystalline form, for example, calcite or aragonite [7].

Oxidation from biocides or radiation induced reactive species is known to damage and change the characteristics of separation membranes. The interaction of reactive oxygen species with membranes is known to produce a great variety of different functional modifications due to direct interaction of the reactive species with the molecular chemistry of the membrane and functional modification of these molecules. Oxidation of membranes may also indirectly impact membrane functions through activation of certain types of ion channels and ion carriers in the membrane. This effect is due to an increase in the polarity of the membrane interior by accumulation of polar oxidation products. The concomitant change of the dielectric constant facilitates the opening of membrane channels and lowers the inner membrane barrier to movement of ions across the membrane. The strong increase of the leak conductance contributes to the depolarization of the membrane potential and can destroy the barrier properties of the membrane

Recently, a new generation of lipid-based membranes, that mimic naturally occurring biological membranes has been developed to purify wastewater. In nature, these types of lipid membranes have the ability to self-repair. The goal of the work conducted in this project is to use synthetic biological techniques to develop a biomimetic membrane system that can self-repair when exposed to chemical and physical damage. The self-repair capability of the biomimetic membrane is derived from the regenerative characteristics of living systems. Ultimately a biomimetic forward osmosis (FO) membrane system will be developed to extend the operating life and improve the performance of membrane based life support systems currently being developed by NASA.

II. Objectives

The objective of this project is to develop a membrane system with extended life capabilities derived from regenerative characteristics of living systems. Initially, this membrane will be designed to function as a water recycling system, but the capability developed will be applicable to a wide range of uses. The final deliverable is a regenerative membrane with metrics that at least meet the stat-of-the-art (SOA) of non-biological water treatment membranes. The success criteria for the SBM project are:

- Demonstration of a functional bio-membrane that can be used for wastewater treatment.
- Demonstration of the ability of the membrane to recover from fouling or oxidative damage.
- Demonstration the membrane has performance characteristics comparable with the current SOA. *

> *SOA - As defined by the ISS Water Processor Assembly (WPA) and other advanced life support technologies such as the Next Generation Life Support – Alternative Water Processor – Forward Osmosis Secondary Treatment (NGLS – AWP – FOST) system*.

III. Technical Approach

The SBM will be integrated into a forward osmosis (FO) purification system to demonstrate its functionality regarding wastewater treatment applications. Forward osmosis membranes are semipermeable membranes able to separate two fluids of different solute concentrations driven by the osmotic potential difference between the two solutions. The membrane allows the solvent to pass through it but rejects the solutes. The flux of solvent across the membrane continues until the osmotic potential across the membrane is equalized and the solute/solvent concentration is in equilibrium. In wastewater treatment applications the feed crosses the semipermeable membrane into the osmotic agent (OA) side (solution with osmotic pressure higher than that on the feed) and the membrane is designed to maximize the flux of water through the membrane while rejecting the contaminants.

Strains of *Escherichia coli* that have been genetically modified to hyper-express fatty acids (FA) and fatty alcohols (FALCs) will be grown in the osmotic agent (OA) loop of the FO membrane system. Fatty acids are a class of lipid that has a single hydrocarbon tail and a carboxyl group. Fatty alcohols have a hydroxyl group in place of the carboxyl group. During growth, the microbes will produce and secrete the desired FA/FALCs in the osmotic agent. The FA/FALCs will diffuse through the membrane from the OA side of the membrane to the wastewater feed-side of the membrane down a concentration gradient forming a fatty acid layer on the feed-side of the membrane where they will serve as a protective agent. The FA/FALC layer will build up until shear forces caused by the circulation of the wastewater feed cause it to shear off. FA/FALC replacement will provide a cleaning of the surface of the membrane.[9, 10]

[9] zNano LLC provisional patent applications 11784391.2, 13/113,930, 2,837,062, 2013-511416, 61/916,617, PCT/US2012/066464, PCT/US2012/066464, PCT/US2012/066464, PCT/US2012/066464

[10] zNano LLC provisional patent application (filed May 2016)

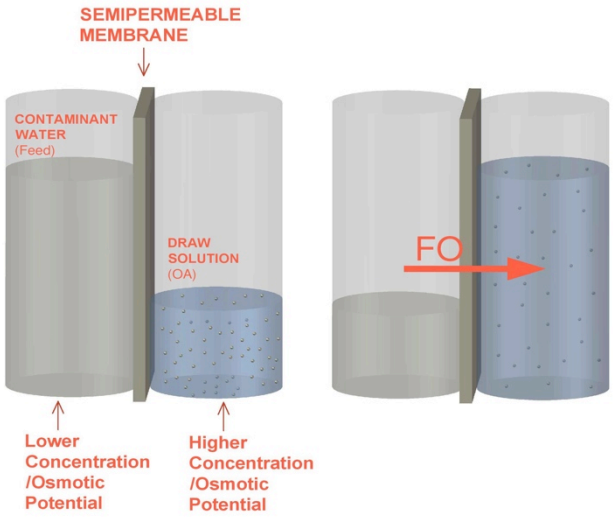


Figure 1. FO diagram illustrating the flow of water through the membrane. *Figure (a) shows two solutions of differing osmotic potential separated by a semipermeable membrane that allows only water to pass from one chamber to the other. As the water moves from the lower concentration fluid (the feed) to the higher concentration fluid (the osmotic agent) the levels in the chambers change indication the flux of water from the feed to the osmotic agent (b).*

Figure 2 provides a simplified drawing of how the bio-membrane will function. As shown, two tail carbon lipids are attached to a porous support forming the foundation of the membrane. It is upon the feed-side of this two-tail carbon lipid bilayer membrane that the FA/FALCs will form a protective regenerating layer. On one side of the support a wastewater feed solution is circulated; on the other side the OA containing the FA/FALC producing organisms is circulated.

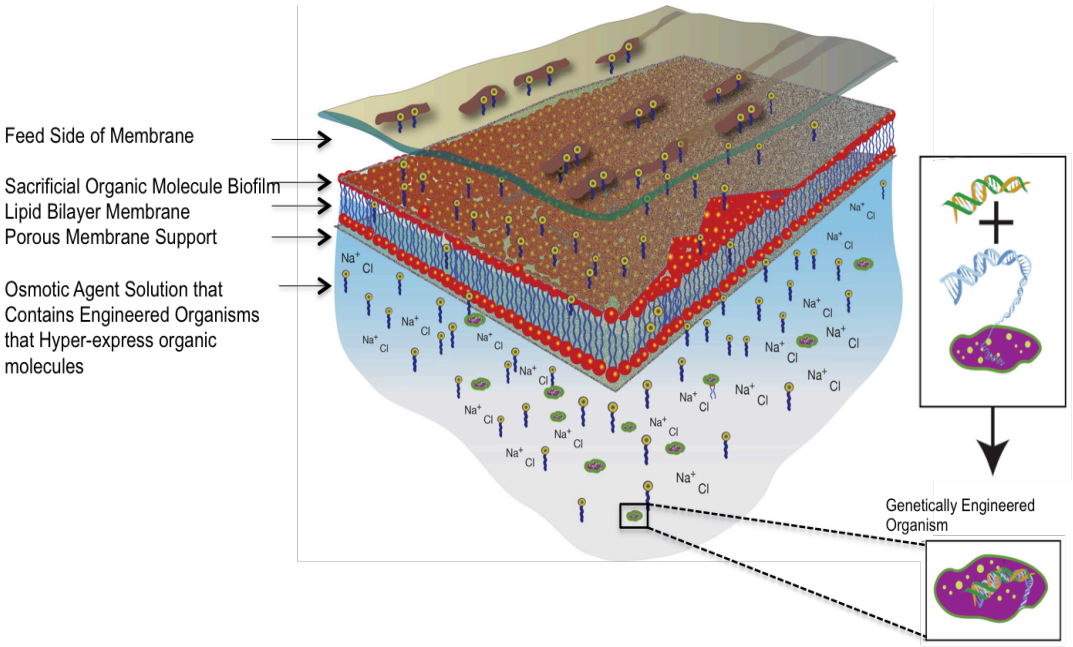


Figure 2. Simplified diagram of structure and function of the proposed SBM. *In this model of the SBM feed passing tangentially across the top surface of the SBM comes in contact with the FA/FACL layer. Minerals and organics which precipitate from the feed as it concentrates bind to the FA/FACL on a layer on the surface of the lipid*

bilayer membrane. This FA/FACL layer is sacrificial and sluffs off do to shear forces of the feed flowing across it. Oxidants and biocides also react with this FA/FACL layer preferentially to the lipid membrane. The FA/FACL are replaced by organisms in the OA that have been engineered to hyper-express them. These organisms produce FA/FACL in the OA. The OA has been modified to increase the solubility of FA/FACL's. The FA/FACL's diffuse back across the SBM from the OA into the feed. In the feed they exceed their solubility limits and precipitate on the surface of the SBM thus replacing the sacrificial layer.

Existing patents and publications describe procedures to engineer microbes to hyper express FA/FALCs, for the manufacture fuels and chemicals. This project will identify various FA/FALCs that can be produced biologically and match them to known membrane characteristics. This will require matching membrane formulations and structures with optimal FA/FALC microbial production. These organisms will then be integrated with several operational configurations of increasing complexity in order to ultimately develop a living membrane that is capable of repairing itself. This integrated system will be constructed to enhance biological stability and operate in a FO water treatment system.

Final Deliverables: At the conclusion of the 3-year project the final deliverable will include the following:

- Functioning bio-membrane test cell designed to demonstrate use in a forward osmosis water treatment system.
- Scientific publications advancing knowledge of bio-membrane systems.
- A technology development plan for future development of the SBM and integration into the NASA developed forward osmosis system.

IV. Fatty Acid Selection

A review of existing literature on FA interaction with calcium and other potential fouling substances identified over 250 functional fatty acids. This list was narrowed down to nine that were the most likely candidate for inclusion into the SBM structure. Target FA molecules were selected with the following properties in mind:

- FA ability to change calcium solubility based on acidity and ability to bind to the SBM.[11, 12]
- FA ability to mitigate organic fouling by partitioning organics into vacuoles.
- FA ability to protect against oxidation through oxidation reaction with biocides and hydroxyl ions.
- FA ability to diffuse into and across the SBM
- FA that can be produced in sufficient quantity in microbial systems and are available through pre-engineered microbes.

Both saturated (no double bonds between carbon atoms) and unsaturated FAs, with a focus on short to medium chain molecules (8-20 carbon atoms) FAs were considered. FAs with even numbers of carbon atoms were targeted because they can easily be produced by enzymatic systems that use the two-carbon building block, acetyl-CoA, through the standard biosynthetic process. The smallest (shortest) fatty acids have the highest diffusion coefficients, through aqueous media, and through the lipid membrane. For all "fouling mitigation" considerations, the smallest FA was chosen. This will help to decrease transport limitations.

The concept used to prevent calcium fouling is to rely on the ability of FA's to acidify an aqueous solution. Then the question is, are the FA's pKa sufficient to achieve acidification. It depends on the FA's and their solubility in the OA. The pKa for fatty acids are ~4.8 . Un-saturation is not helpful because it drives the pKa in the wrong direction. The solubility of FA's is strongly dependent on carbon-chain length. Several fatty acids were examined for their suitability for the membrane and to correct calcium fouling. The top 3 choices for this function are octanoic acid, hexanoic acid, and decanoic acid.

The organic fouling of a membrane is caused by a deposition of biopolymers. Previous studies have demonstrated that the major percentage of those polymers are proteins and polysaccharides (Zhou et al., 2007), more

[11] zNano LLC provisional patent applications 11784391.2, 13/113,930, 2,837,062, 2013-511416, 61/916,617, PCT/US2012/066464, PCT/US2012/066464, PCT/US2012/066464, PCT/US2012/066464

[12] zNano LLC provisional patent application (filed May 2016)

specifically polysaccharides and other non-setteable organic matter with a molecular weight larger than 120 000 Da (Rosenberger et al., 2006). Biological precipitation can be another contribution to inorganic fouling. The biopolymers contain ionisable groups (COO^-, CO_3^{2-}, SO_4^{2-}, PO_4^{3-} and OH^-) which are easily capturable by metal ions. Metal ions play a significant role in the formation of fouling layers, which can bridge the deposited cells and biopolymers and then form a dense cake layer. There exists a synergistic interaction among biofouling, organic fouling and inorganic fouling (Meng et al., 2009). [13, 14]

For urine containing feeds the most likely "problem protein" may be blood albumin (MW 68kD). Possible mechanisms of fatty acid mitigation of protein fouling include, fatty acid complex formation with proteins. This is a complex process, only well-understood for palmitic acid (16C, no double bonds). Testing competed in our lab using membranes fouled with a bioprocessed feed of urine + hygiene + humidity condensate indicated a polyamide spectrally similar to liquid pyrolyzate of melamine-urea-formaldehyde resin. From what we know about fatty acid binding to proteins, longer chain fatty acids work best. Unsaturation does not seem to help. The top 3 choices are palmitic acid (16C, saturated), myristic acid (14C, saturated), and lauric acid (12C, saturated).

The biomembrane may become oxidized/damaged by reactive oxygen species including hydroxyl radicals, superoxide anions, and hypochlorite anions that cause lipid "peroxidation." This results in a lipid radical, followed by the formation of a lipid with an attached peroxy group. This occurs at sites of unsaturation in lipids, that is, it requires a double bond. The basis for choosing one unsaturated fatty acid over another is not entirely clear. As part of this project experimental work will be conducted to evaluate how fatty acids prevent membrane-lipid oxidation. However, for now the top 3 choices are oleic acid, linoleic acid and alpha-linoleic acid.

The current list of potential candidate fatty acids will be examined and narrowed to a few candidate FA/FALCs that are produced by available engineered organisms. To accomplish this, we will obtain target organisms capable of producing the candidate FA/FALCs via contract and define the OA parameters that allow the targeted organisms (strains of *E. coli* producing FA/FALCs) to grow well and produce the needed FA/FALCs in the OA. This will be accompanied by a determination of the target organisms' stability in the OA relative to the life of the membrane, as well as their potential for biofilm formation, and potential bio-fouling.

A protocol has been developed to quantify the diffusivity of FA/FALCs into and through the membrane to the feed-side to form a layer. Concomitantly, we will demonstrate the fabrication of bio-membranes with structural lipids and fusion FA/FALCs leading to the production of a set of membranes with a range of FA/FALC permeation rates.

The manufactured membranes will be evaluated to determine their performance and ability to meet the Key Performance Parameters (KPPs). This will include an evaluation of the ability of the membrane to self-repair by building a protective layer of FA/FALCs that interact with calcium, targeted organics, and targeted oxidants. As the FA/FALCs interact with these chemicals, they will be removed from the membrane by shear forces caused by the feed flowing across the membrane. The older FA/FALCs will be replaced by newer FA/FALCs that will permeate through the membrane from the OA, where they exist in high concentrations [13, 14]

V. Engineered Organism

The genetic engineering approach used is to collaborate with commercial suppliers of pre-engineered organisms in order to acquire engineered strains of *Escherichia coli* that can hyper-express the desired fatty acids. Within nature, fatty acid biosynthesis is responsible for cell structure and energy storage. Figure 3 provides a structural diagram of a fatty acid containing a carboxylic head group and an 18 carbon hydrocarbon chain tail structure. Genetic engineering provides the ability to produce a wide range of fatty acids with differing head groups and chain lengths. This highly efficient pathway when genetically linked to product pathways provide unique catalysts that enable simple processes to produce fatty acids and alcohols.

[13] zNano LLC provisional patent applications 11784391.2, 13/113,930, 2,837,062, 2013-511416, 61/916,617, PCT/US2012/066464, PCT/US2012/066464, PCT/US2012/066464, PCT/US2012/066464

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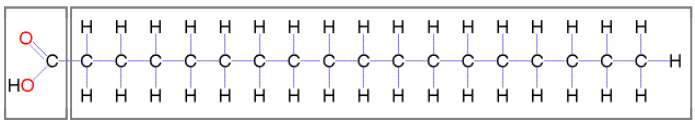


Figure 3. **Molecular composition of fatty acids.** *This is an example of a saturated fatty acid containing 18 carbon atoms and no double bonds, but many different structures are possible.*

These engineered organisms will be grown in a multi-culture reactor in the FO OA loop that produces multiple fatty acids. The engineered organisms will modified to enhance the catalytic capabilities of microbial metabolism. Unique sequences of DNA enable chemical transformations in a cell by harnessing the power of enzymes. A "metabolic pathway" is composed of a set of these enzymes working together to sequentially convert one compound in the cell (feedstock), through a series of selective chemical transformations, to a different compound. There are several identified commercial entities that have developed the most efficient industrial biotechnology method for the synthesis of hydrocarbon chains – fatty acid biosynthesis – and are combining is discovery with new engineered biochemical pathways that convert a wide range of feedstock's into FA and FALC's. By engineering all of the catalytic steps into a single microbial catalyst these suppliers have enabled a simple, selective conversion processes to direct targeted products.

One of the key patents in this field (application number 20130035513) addresses a key regulator in the initiation of fatty acid biosynthesis known as phosphopantetheinyl transferase. This enzyme is responsible for transferring 4`-phosphopantetheine from coenzyme A to a conserved serine residue on acyl carrier protein, which is responsible for shuttling around 4`-phosphopantetheine. This pathway is essential for the functioning of the Fatty Acid Synthase enzyme.

It should be noted that the majority of studies of microbes, including *E. coli,* when grown in microgravity conditions reported changes in growth kinetics and biomass [8] . Specifically, a reduced lag phase and increased final cell populations in space. It appears that cell motility may be the key variable responsible for the seemingly disparate results. As cell mobility increases kinetics approaches 1 g levels.

VI. Nutrient Balance

Because *E. coli* are heterotrophic, they require a source of nutrients that may not be met by the nutrients (organic, nitrogen, phosphorous and trace mineral compounds) present in the feed side and that diffuse through to the OA side of the membrane. Figure 4 shows a relative growth curve for wild type E. coli grown in simulate OA. In this Figure, cultures of *E. coli* K-12 were grown in a plate reader on osmotic agent with various nutritional supplements at 37°C for 16 hours with the OD_{600} measured every 5 minutes.

Figure 3 show that osmotic agent generated using the "leaky" SBM membranes allows for modest growth of *E. coli* with no supplementation, though growth rates and yields are lower than on standard M9 medium. In these experiments, organic carbon appears to be a limiting factor for growth among the nutrients tested. These data also show that the osmotic agent generated with "leaky" membranes has a growth-inhibitory effect when added to M9 medium.

To investigate this potential requirement for supplying or resupplying these nutrients we will evaluate the transport of metabolic requirements across the bio-membrane and their ability to support *E. coli* growth. If the metabolic requirements are not met by transport across the membrane from the feed-side to the OA side of the membrane a study to determine if the addition of marine cyanobacteria (photoautotrophs) that produce and excrete sugars to the OA side of the system will alleviate a resupply requirement. Additionally, a study to determine how the cyanobacterial growth requirements (lighting and salt water) will fit into the system will be conducted. Finally, a trade study will be conducted to determine the best path forward to minimize any resupply requirements (e.g., equipment, mass, power, and volume).

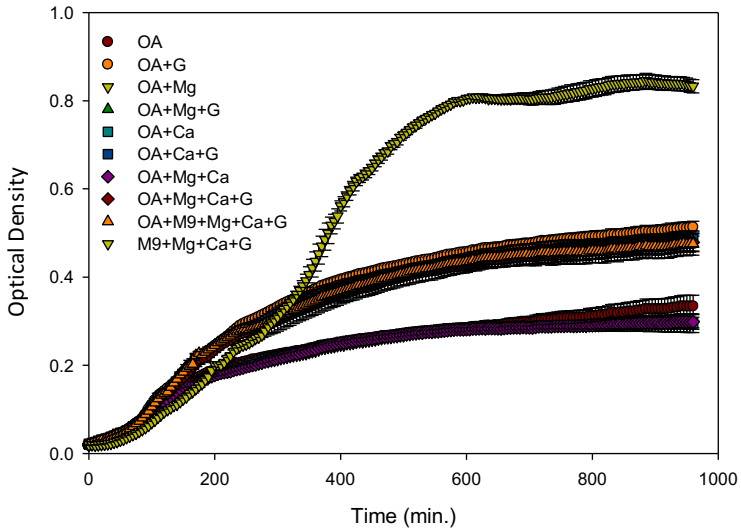


Figure 4. Growth curves for osmotic agent from "leaky" membranes; *effects of glucose, $MgSO_4$ and $CaCl_2$ addition were investigated. Cultures were grown in osmotic agent with varying components of M9 medium added. G is glucose, Mg is $MgSO_4$, Ca is $CaCl_2$, and M9 is complete M9 glucose medium.*

VII. Key Performance Requirements

The following requirements have been assumed in order to direct the technology development tasks defined in this work plan. These requirements will be continually updated as future missions mature. Table 1 documents Key Performance Parameters (KPPs) for all the technologies developed under the project. KPPs will be evaluated either through analysis or testing.

Table 1. KPPs for SBM project

KPP	Units	SOA	Threshold	Goal
Water flux rate	$L/M^2/H$	2	4	22
Organic rejection	%	50	50	95
Inorganic rejection	%	98	96	99
Ammonia/ammonium rejection	%	50	85	95
Water recovery ratio	%	75	85	87
Water savings, ISS model	Kg	NA	329	624
Mass savings over SOA	Kg	NA	352	1000

KPP Assumptions:

- Water Flux Rate: Threshold based on commercially available membranes and Goal based on best commercial Lipid membrane.
- Organic rejection: Threshold based on commercially available membrane and Goal based on best commercial Lipid membrane.

- Inorganic rejection: Threshold based on commercially available membrane and Goal based on best commercial Lipid membrane.
- Ammonia: Threshold based on commercially available membrane and Goal based on best commercial Lipid membrane.
- Water recovery ratio: Threshold based on ISS target of 85% for Urine Processing Assembly. The Goal is a predicted maximum recovery ratio possible for the UPA urine feed.
- Water savings, ISS model: Threshold based on water savings generated from increasing water recovery ratio from 74% to 85%. This is the current ISS operational objective. The Goal is based on increasing the water recovery from 74% to 87% max.
- Mass savings over SOA: Threshold SOA is defined as the FOST system. The savings is the sum of water savings and membrane replacement savings using a urine + flush feed. The Goal is based on operation at a higher water recovery ratio. Membrane replacement calculated bases on 5.2 Kg per replacement and 2000 hr.

The project will advance the SBM technology from TRL 2 to TRL4. During the first year it was determined that the original approach to produce and harvest lipids as the active layer of the membrane was impractical and a new approach that employed FA/FALCs was implemented. In year 2, proof of concept work will be completed resulting in TRL 3. In the final year, the system will be validated in the laboratory environment when the FA/FALCs from engineered organisms are integrated with the membrane matrix and the resulting bio-membrane test is evaluated in in a FO environment.

Table 2. TRL advancement

Year	TRL Level	**Rational**
1	2	Complete applied research for engineering organisms and evaluate the FA/FALCs
2	3	Complete proof of concept work- confirmed ability to produce membrane using biologically relevant FA/FALCs
3	4	Validation of the system in laboratory environment – organisms producing FA/FALCs to repair the membrane evaluated in a FO test environment.

VIII. Conclusions

The final deliverable of this project is a functioning test stand that is capable of demonstrating the re-generable characteristics of the SBM. A simplified flow diagram of this system is provided in Figure 5. The system will use urine as the feed. The urine will be re-circulated through the membrane contactor across the SBM using a recirculation pump. On the back side of the membrane contactor the OA will be re-circulated using a similar pump. The genetically engineered organisms will be grown on the OA and the fatty acids produced will permeate back through the SBM. The OA will be modified to increase the solubility of the fatty acids. As the fatty acids pass through the SBM and come in contact with the feed they will become subject to the normal solubility limits and precipitate on the surface of the membrane. This fatty acid layer will serve as a sacrificial layer on the feed side of the membrane. Nutrients fro growth will at least partially permeate from the feed into the OA. Options will include integration of a cyanobacteria photo bioreactor in the OA feed reservoir to supplement the organic content of the feed by producing sugars and gas exchange contactors in the recirculating OA to control O_2 and CO_2. The system will also be provided with an OA regeneration system that facilitate long term continuous operation of the system to conduct extended life studies.

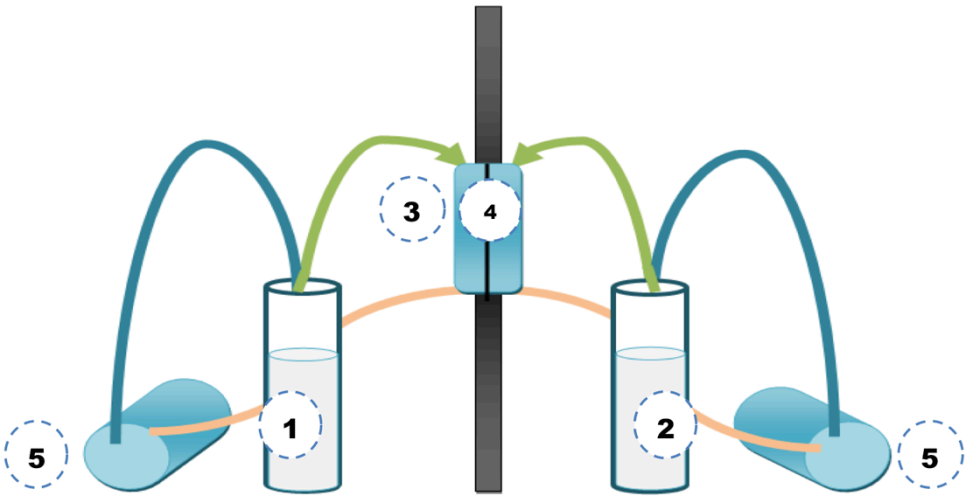


Figure 5. Simplified FO testing cell diagram. *The SBM membrane (4) will be placed in the membrane contactor which is composed of two acrylic plates bolted together (3); the active layer of the membrane facing the feed which is circulated through reservoir (1) and the membrane support facing the osmotic solution which is circulated through the reservoir (2). Gear pumps (5) will be used to recirculate the feed (1) and OA (2) through the membrane contactor (3) and across the SBM membrane (4).*

By the end of year 2 the project will have:

- Obtained targeted FA/FALC producing organisms and tested their ability to produce FA/FALCs in the OA.
- Confirmed the ability to fabricate a membrane with the targeted biologically produced FA/FALCs.
- Conducted a trade study to determine best path forward to minimize any resupply requirements (e.g., equipment, mass, power, and volume).
- Determined the best method to deliver and control oxygen and remove and control carbon dioxide in the OA.
- Completed an evaluation of the manufactured membrane.

By the end of Year 3 the Project will have:

- Balance FA/FALC production rate in the OA loop with diffusivity and their calculated demand by the membrane for cleaning wastewater.
- Construct integrated SBM system
- Test performance of integrated SBM system.

IX. References

1. Drews, A. (2010). Membrane fouling in membrane bioreactors—characterization, contradictions, cause and cures. *Journal of Membrane Science*, *363*(1), 1-28.
2. Jun, Z., YANG, F. L., MENG, F. G., Peng, A. N., & Di, W. A. N. G. (2007). Comparison of membrane fouling during short-term filtration of aerobic granular sludge and activated sludge. *Journal of Environmental Sciences*, *19*(11), 1281-1286.

3. Rosenberger, S., Evenblij, H., Te Poele, S., Wintgens, T., & Laabs, C. (2005). The importance of liquid phase analyses to understand fouling in membrane assisted activated sludge processes—six case studies of different European research groups. *Journal of Membrane Science*, *263*(1), 113-126.
4. Meng, F., Chae, S. R., Drews, A., Kraume, M., Shin, H. S., & Yang, F. (2009). Recent advances in membrane bioreactors (MBRs): membrane fouling and membrane material. *Water research*, *43*(6), 1489-1512.
5. Siobhán Francesca E. Boerlage, "Scaling and Particulate Fouling in Membrane Filtration Systems", Taylor & Francis; 2001, ISBN 90-5809-242-9
6. T.R. Bott, "Fouling of Heat Exchangers (Chemical Engineering Monographs)", Elsevier Science, 1995.
7. "Modeling PWR Fuel Corrosion Product Deposition and Growth Processes (5)", Technical Report 1009734, Electric Power Research Institute, Palo Alto, California, USA, 2004.
8. Horneck, G., Klaus, D., & Mancinelli, R. L. 2010. Space Microbiology. Microbiol. Molec. Biol. Rev. 74:121-156.